

# Yusef Syed

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## Education

### University of Toronto

Expected May 2030

First-year Mathematical & Physical Sciences, Faculty of Arts & Science

### Mathematics and Computer Science (Intended Double Major)

Current coursework: programming, calculus with proofs, linear algebra, statistical reasoning.

Prior coursework: Data Structures, Lehigh Carbon Community College (dual enrollment).

## Research Projects

*Personal projects with AI-assisted implementation; code, methods and limitations are linked.*

### Agent Eval Mutation Lab

Aug 2026 - Present

Python, Inspect, Docker, SQLite | [Code and evidence](#)

- Built a resumable evaluation engine separating proposed actions, executed tool effects and unknown outcomes; **104 scorer configurations across 13 synthetic scenarios**.
- Integrated Docker-isolated Inspect tools with independent database-effect checks; tested commit-before-error and later-revocation faults, plus **24 local-model tool-use samples**.
- Preregistered a 624-trial two-model study; reported **5 of 6 failed validity gates**, withheld comparative model claims and documented missing-output sensitivity.

### Tiraz Garment Completion Study

Aug 2026

Python, PyTorch, NumPy | [Code and results](#)

- Trained three seeded set encoders on **27,683 Fashionpedia annotation groups**; achieved **58.5-59.0%** top-1 versus a **52.7%** co-occurrence baseline on 962 held-out source groups.
- Separated training, tuning and calibration groups; a context-removal stress test reduced nominal 90% prediction-set coverage to **55.6-57.9%**. Annotation-only study, not image recognition.

### ShiftProof

Sep 2026

Python, local LLM, constraint verification | [Code and evaluation](#)

- Built a local volunteer-scheduling prototype that checks model proposals with an independent verifier and keeps apply, approve and export actions under coordinator control; validated a synthetic workflow and inspected four model requests for free-text isolation.

## Open-Source Contributions

### Contributor

Aug 2026 - Present

- Contributed **9 merged upstream PRs** across Microsoft Agent Lightning, Microsoft PyRIT, W&B RAI Toolkit, Meridian Labs Inspect Scout and Kornia. [Merged contributions](#)
- Fixed asynchronous child scoring and unassessed-result handling in W&B's [CompositeScorer](#); added focused regressions for aggregation and async behavior.
- **Open PyTorch PR (AI-assisted)**: implemented CPU/CUDA shape gradients for regularized incomplete gamma. Local source-build checks compared **3,486 derivative outputs** across two CPU builds and RTX A4000 CUDA, with no corpus failures. [PR](#) | [Validation](#)

## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL

**Research and testing:** PyTorch, NumPy, Inspect, pytest, mypy, Git, GitHub Actions

**Systems:** Docker, SQLite, PostgreSQL; C++/CUDA contribution experience

## Leadership

**Feeding the Valley:** Vice President (2025-2026) | **FIRST Robotics Team 834:** Fundraising Lead (2022-2026)